DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING

Experiment No. 9

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Branch: BE CSE
Section/Group: KRG 2B
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Subject Code: 23CSH-314
Subject Name: System Design
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1. Aim:

To design a highly scalable ticket booking platform that allows users to discover and purchase tickets for entertainment events, ensuring high availability for event discovery and strict consistency for seat reservations.

2. Objective

- Understand the architecture of a massive-scale ticketing and reservation system.
- Identify functional requirements such as event discovery, real-time seat mapping, and booking execution.
- Identify non-functional requirements including handling 100 Million DAU, search availability, and transaction consistency.
- Analyze the decoupling of payment workflows and status updates using event-driven stream processing (Kafka).
- Design REST APIs for event metadata, seat layouts, and secure transaction processing.

3. Procedure

- Studied real-world ticket booking architectures and concurrent transaction handling.
- Identified core entities such as Users, Events, Venues, Shows, Seats, and Bookings.
- Analyzed TTL-based distributed locking mechanisms to temporarily hold seats and prevent double-booking.
- Collected functional and non-functional requirements for event discovery and high-demand ticket sales.
- Designed APIs for search filtering, comprehensive event display, and booking execution.
- Evaluated CQRS-like patterns by separating read-heavy search operations (via Elasticsearch and CDC pipelines) from write-heavy primary databases.
- Analyzed asynchronous message brokers (Apache Kafka) for handling third-party payment callbacks, seat status updates, and user notifications.

4. System Requirements:

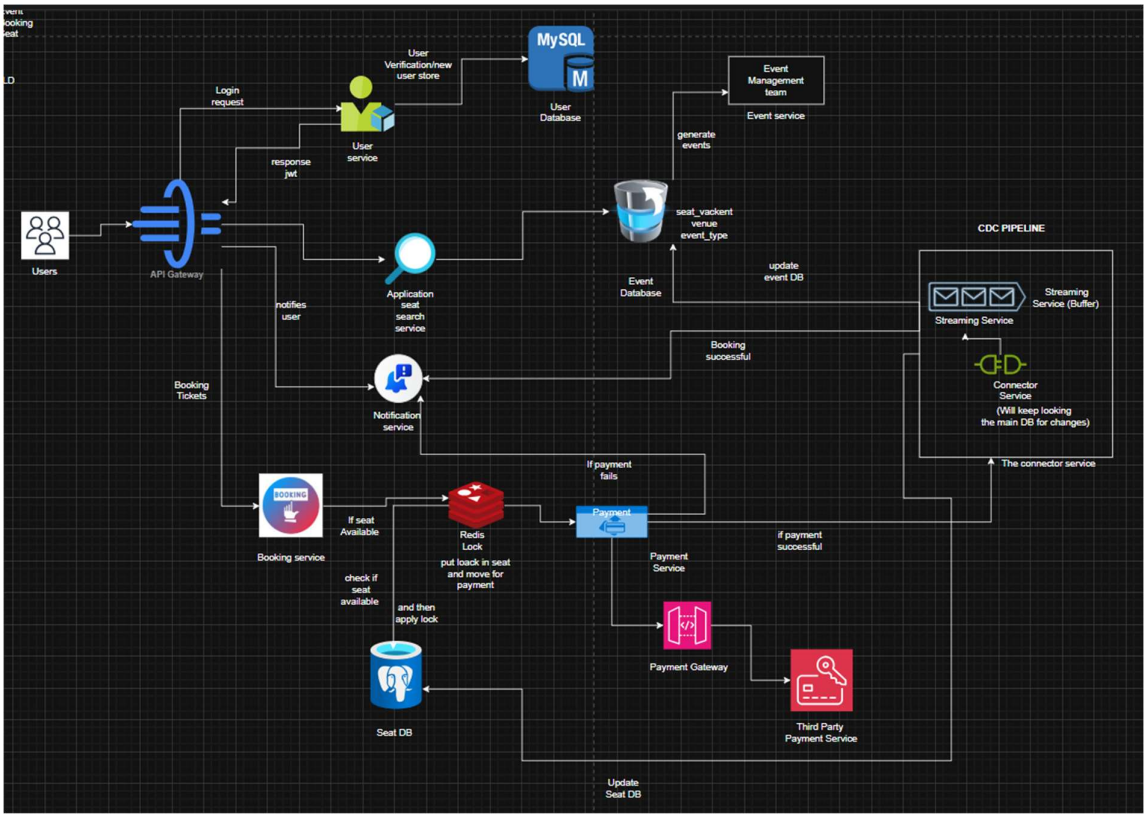
A. Functional Requirements

- The system must allow users to search and discover events using filters such as event title, location (city), and date.
- The system must display detailed event information including seat availability (available vs occupied).
- The system must provide additional event metadata such as description, cast, duration, language, and ratings.
- The system must allow users to select specific seats from a seating layout.
- The system must enable users to securely complete ticket booking transactions and receive confirmation.

B. Non-Functional Requirements

- The system must support scalability to handle up to 100 million daily active users (DAU).
- The system must ensure high availability for search and event viewing, even under heavy traffic.
- The system should tolerate slight data staleness (event listings, seat counts) instead of system downtime.
- The system must ensure strong consistency during seat booking to prevent double booking.
- The system must provide low latency responses for search, viewing, and booking operations.

5. High Level Design (HLD):



6. Learning Outcomes (What I Have Learnt)

- Designed a highly scalable, strictly consistent ticket booking architecture.
- Identified critical functional and non-functional requirements for high-demand event ticketing.
- Designed REST APIs and decoupled asynchronous processing pipelines for payments and notifications.
- Understood the application of Elasticsearch for fast read-heavy queries, TTL-based database locks for concurrency control, and Kafka for robust event-driven workflows.